Preliminary paleomagnetic results from the Permian Asitka Group, British Columbia[1]

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Tuffs, rhyolites, and basalts collected at seven sites (25 samples) in the Early Permian Asitka Group of Stikinia have similar paleomagnetic inclinations, and corresponding paleolatitudes of 23 ± 6, 21 ± 4, and 23 ± 5° respectively. Both polarities occur, so the paleolatitudes could be either north or south of the paleoequator, although the polarity bias favours the former option. The paleolatitude expected for the sampling region is 27 ± 4°N. Adopting the paleomagnetically more probable northern option, the mean displacement relative to the North American craton is 5 ± 6°, indicating that Stikinia was in much the same latitudinal position relative to the craton during the Early Permian as it is at present, although possibly somewhat to the south. The declinations are in poor agreement, indicating that relative rotations occurred in this faulted terrane.

Les 25 échantillons de tufs, rhyolites et basaltes prélevés à sept sites dans le groupe d'Asitka, d'âge permien inférieur, de la Stikinia possèdent des inclinaisons magnétiques similaires, et dont les paléolatitudes correspondantes sont 23 ± 6°, 21 ± 4° et 23 ± 5°, respectivement. Les deux polarités coexistent et par conséquent les paléolatitudes peuvent être aussi bien au nord qu'au sud du paléoéquateur, cependant la tendance des polarités favorise l'hémisphère nord. La paléolatitude calculée pour la région qui a été échantillonnée est 27 ± 4°N. Si on accepte que la probabilité est plus élevée pour une paléolatitude située dans l'hémisphère nord, le déplacement moyen relatif du craton de l'Amérique du Nord est de 5 ± 6°, indiquant que la Strikinia occupait sensiblement cette position de latitude par rapport au craton durant le Permien inférieur, tout comme actuellement, bien qu'elle aurait pu être située un peu plus au sud. Les déclinaisons magnétiques varient considérablement, et elles témoignent que des rotations relatives se sont produites dans ce terrane faillé.

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Geology

The eastern part of the extensive terrane in northwestern British Columbia called Stikinia (Fig. 1) comprises the Lower Permian Asitka Group, the Upper Triassic Takla Group, and the Lower to Middle Jurassic Hazelton Group. These rocks are overlain by Middle Jurassic to Lower Cretaceous clastic sedimentary rocks of the Bowser Basin, which were in part derived from the Cache Creek terrane. The Asitka Group has been described by Lord (1948) and Monger (1977). The most complete section lies about 3 km east of Dewar Peak, in the McConnell Creek map area (56°44′N, 126°49′W). There it forms part of the uplifted rim of upper Paleozoic and lower Mesozoic strata that surrounds the Bowser Basin (BB of Fig. 1). The Early Permian (late Sakmarian to early Artinskian) age of the Asitka Group is well established (Ross and Monger 1978; Rigby 1973). It was deposited about the middle of the Kiaman superchron, which extends from the Westphalian to the end of the Paleozoic. During the Kiaman, the geomagnetic field was predominantly of reversed polarity.

In the Dewar Peak area, the Asitka Group consists of three generally southward-dipping units exposed in a faulted succession (Fig. 2). The lowest unit, whose base is faulted against Lower Jurassic rocks of the Hazelton Group to the east, comprises up to 600 m of basalt, argillite, chert, and carbonate. The middle unit consists of about 1500 m of basalt flows overlain by red flow-banded rhyolite and prophyry with intercalated breccia. The upper unit, a few hundred metres thick, comprises basalt, basaltic tuff, pink tuffaceous limestone, and chert. The Asitka Group is overlain stratigraphically by upper Karnian argillite and volcaniclastics of the Dewar Formation, which is part of the Upper Triassic Takla Group (Monger 1977, Fig. 2).

Logistical restrictions limited the extent of sampling. Since there will be no opportunity to sample further in the near future, and since the rocks possess magnetizations that are among the most stable yet observed in the Cordillera, we are putting the data on record despite its limited extent. Collections comprise 15 hand samples of basalt from five sites spanning 300 m, 5 hand samples of rhyolite spanning 10 m from the middle unit, and 5 hand samples of grey tuff from one site spanning 2 m from the upper unit (Fig. 2). Samples were cored in the laboratory and one or two specimens cut from each core. Bedding attitudes are considered to be accurate to better than 5°.

Results

The *basalt* flows are, for the most part, fine-grained purple or reddish in colour and amygdaloidal. They contain abundant finely disseminated magnetite and hematite and occasional blocky magnetite grains 10 μm in diameter with ilmenite intergrowths—a highly oxidized assemblage. About one-half of the specimens have very stable single-component magnetizations. Their unblocking temperatures fall in a narrow range, sometimes close to that for magnetite, sometimes close to that for hematite (Fig. 3). Demagnetization in alternating fields (af's) up to 125 mT yielded no change in direction and the intensity drops by about one-half. Such specimens respond well to thermal cleaning in the range 300–600°C, and for uniformity a "cleaning" temperature of 550°C was used. Data obtained in this "direct" way are shown by open circles in the lower

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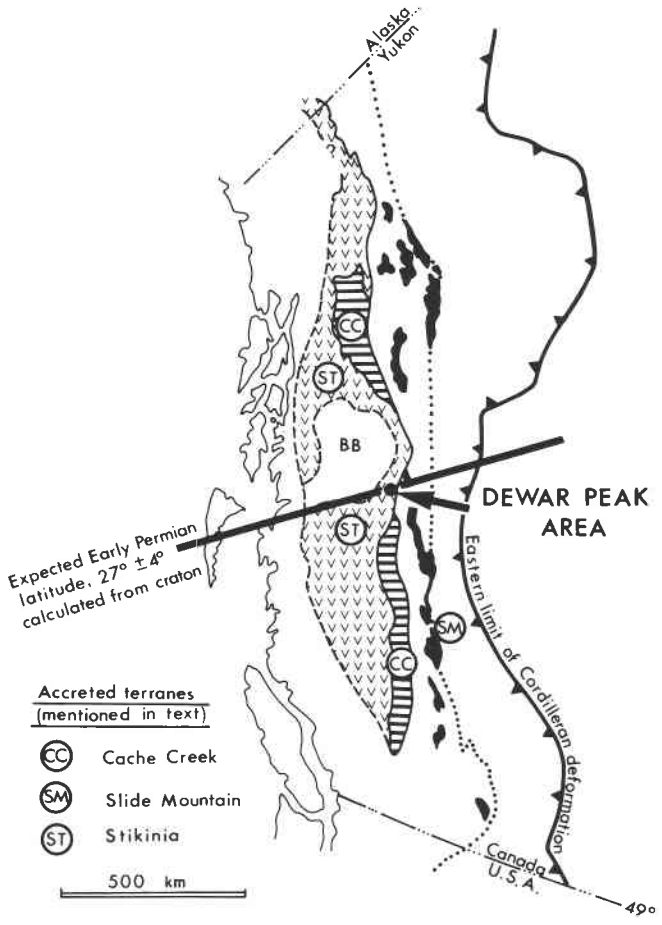


FIG. 1. Location of Dewar Peak in relation to some accreted terranes of the Canadian Cordillera. The Slide Mountain terrane, SM, is the easternmost of the accreted terranes. Stikinia, ST, lies west of Cache Creek with its exotic Tethyan faunas. BB is the Bowser Basin. The line of paleolatitude through the sampling area is that expected from cratonic data (27 ± 4°). The observed paleolatitude at Dewar Peak is about 22° (see Table 1 for details). The probable eastern limit of accreted terranes is shown by the dotted line.

half of Fig. 4. After correction for tilt of the beds, the magnetizations point southeast with negative inclination (Table 1).

The other basalt specimens have stable but complex magnetizations. In the unusual example shown in Fig. 5, the intensity during thermal demagnetization falls suddenly above 500°C, then rises, and falls again. The directions are well grouped up to 500°C, and then they show irregular, but not large, changes up to 670°C (Fig. 6). Up to 450°C the orthogonal plots show little change in direction. This is followed by a rapid movement, first towards, then away from, and finally towards the origin again, and the temperature increases (Fig. 7). The suggested structure of this complex magnetization is shown schematically in Fig. 8. The first magnetization (MG) to be removed has the unblocking temperature of magnetite. The second (HT1) has a somewhat higher unblocking temperature and is carried perhaps by fine-grained or impure hematite. The last (HT2) has the unblocking temperature of pure hematite. MG and HT2 have negative inclinations, like all other basalt magnetizations, but HT1 has positive inclination and is reversed from them. HT1 is a normal magnetization embedded in an otherwise reversely magnetized specimen within a reversely magnetized basalt sequence (Fig. 4). It is unlikely, therefore, that the HT1 magnetization records a reversal of the geomagnetic field that occurred as the lavas initially cooled. As we describe below, the overlying tuffs have directions similar to HT1. Hence we suggest that HT1 was probably acquired chemically during burial and diagenesis, at a time when the overlying tuffs were being deposited; that is, it was acquired in Asitka time during a brief interval shortly after the lava was irrupted when the field had normal polarity. It is noteworthy that short intervals of normal polarity have been recorded within the Kiaman at about this time (Irving and Pullaiah 1976). In the exceptional specimen of Figs. 5–7, the unblocking temperatures for each of the three magnetizations do not overlap appreciably, and the removed magnetizations, calculated for the temperature intervals during which rapid changes occurred, provide good estimates of the directions of all three magnetizations. In most other specimens, however, the unblocking temperatures of two of the magnetizations overlap, and only one magnetization can be satisfactorily isolated. Such "indirect" estimates of directions of magnetization (directions of removed magnetizations) are plotted as open diamonds in the lower half of Fig. 4. The means of all "direct" and "indirect" estimates are statistically identical (Table 1). Experiments with af demagnetization up to 125 mT were also carried out, but they did not allow satisfactory separations to be made of the magnetizations present.

The *rhyolites* are red and very fine grained. They are magnetically simple. Thermal decay curves show first an increase, as a small magnetization close to the present field is being removed (Fig. 9). Alternating field demagnetization up to 125 mT showed a similar small increase of 10–20%. At higher temperatures and fields the orthogonal plots march directly to the origin, showing that above 300°C only one magnetization is present (Fig. 10). Unblocking temperatures are mainly between 600 and 700°C mainly. Presumably, hematite is the carrier. Polished sections reveal disseminated bireflective grains, too small to be identified microscopically. The directions are well grouped and point east with negative inclination (Fig. 4).

The *tuffs* studied are calcareous and contain pumice fragments. They have simple, but very weak, magnetizations. Heating first removes a magnetization directed close to the present field (Fig. 11). Thereafter, the magnetization remains little changed in direction up to 650°C, although the directions are somewhat scattered, because the intensities are weak (the intensity after heating to 650°C and cooling in zero field is less than 10^{-4} A·m^{-1}). Above 150°C, the orthogonal plots decay linearly to the origin, with some random scatter, largely instrumental in origin (Fig. 12). Alternating field demagnetization up to 100 mT yields little change in direction. Presumably, hematite is the carrier of the magnetization. Disseminated reflecting particles, too small to be identifiable, were observed in polished sections.

Discussion

The inclinations relative to paleohorizontal are in excellent agreement, the means for the three units differing by only 2°, well within error (Table 1). Before corrections for dip of the beds are made, the inclinations are 29° (tuff), 51° (rhyolite), and 35° (basalt), and are in poor agreement. Evidently the magnetizations predate tilting. The date of tilting is not known precisely but is pre-Tertiary. The magnetizations present are very stable. Coercive forces are always high, exceeding 100 mT. Unblocking temperatures are above 500°C and

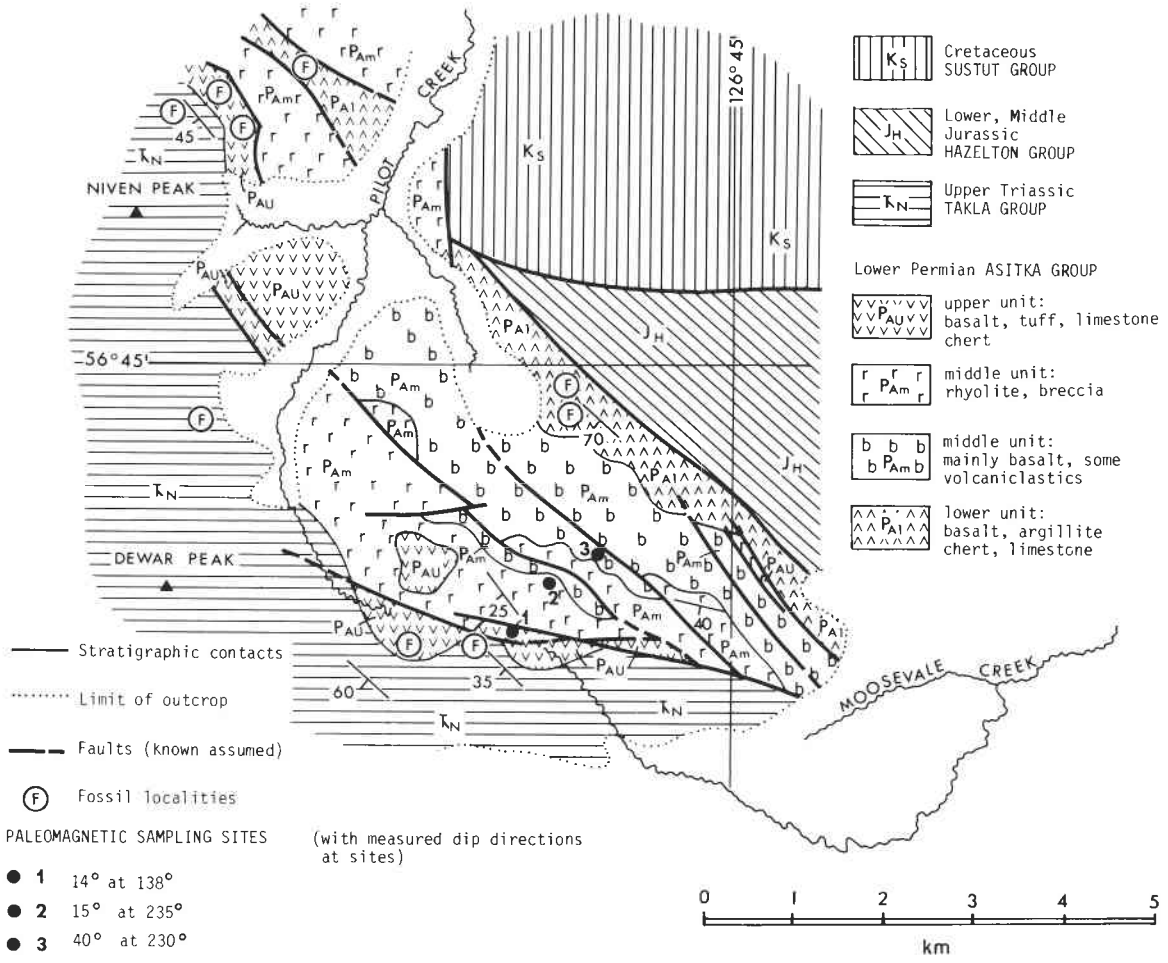


FIG. 2. Sketch map of Dewar Peak area, McConnell Creek (94D) map area showing sampling localities; 1 tuff, 2 rhyolite, 3 basalt (from Monger 1977).

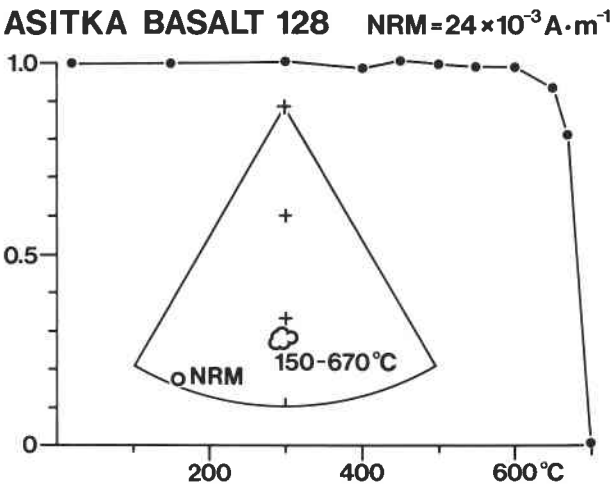


FIG. 3. Thermal demagnetization of basalt showing single component magnetization. Changes in intensity and direction (relative to present horizontal) are shown.

commonly above 650°C—temperature far higher than those the rocks have experienced subsequently. The directions of magnetization (Fig. 4) are well removed from directions of known later overprinting, and studies of thin sections show no evidence of secondary alteration. Hence there is no evidence that the magnetic record has been overprinted. For the above reasons, we suggest that the magnetizations record the geomagnetic field during the formation of the Asitka Group.

The mean inclinations after correction for tilt correspond to paleolatitudes of 21−23°, with errors varying from 4 to 6° (Table 1). The inclination itself does not tell us whether the rocks were magnetized north or south of the paleoequator, but the polarities could shed some light on this ambiguity, because during the Early Permian the geomagnetic field was reversed for about 95% of the time (Irving and Pullaiah 1976). The inclinations are directed predominantly upwards (negative), which is the polarity expected for a northern hemisphere position in a geomagnetic field of reversed polarity. Downward (positive) inclinations occur in the tuff (Fig. 4), as well as in a rare component in the basalts (HT1 of Figs. 5−7). Based on a northern hemisphere position, the polarity bias, measured as a ratio of normal to reversed sites, is 1N:6R. The bias, although marked, is less strong than that expected from a truly random sample of rocks of Kiaman age. Nevertheless, a northern rather than a southern hemisphere position is favoured.

If the northern hemisphere position is adopted, the paleolatitudes (21−23°, Table 1) are in reasonably good agreement with the range expected (27 ± 4°) for this part of British Columbia (Fig. 1); the mean relative paleolatitudinal displacement is 5 ± 6° from the south, which is not significant as 95%

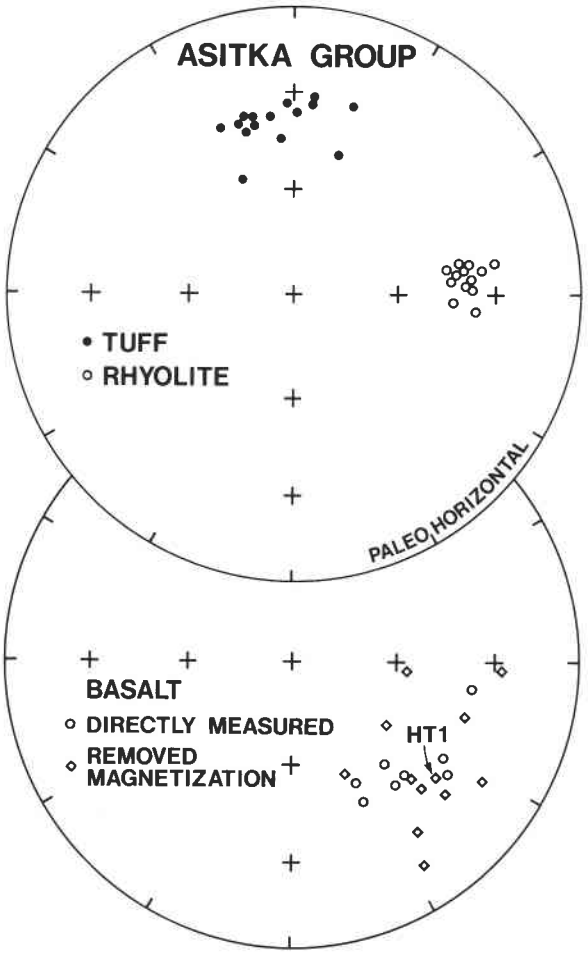


FIG. 4. Thermally cleaned directions relative to paleohorizontal. Bedding altitudes are given in the notes to Table 1. In the upper diagram, solid (open) symbols denote downward (upward) inclinations. Below, the inclinations are all upward except for HT1. HT1 is the direction of the lower blocking temperature hematite magnetization of Fig. 6, which has here been reversed to allow easier comparison.

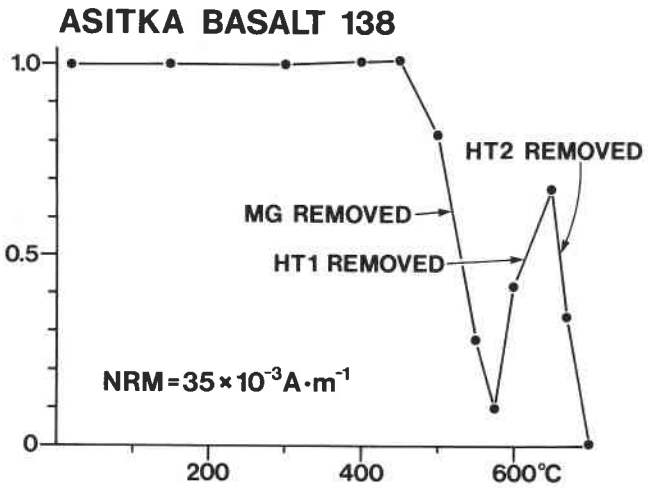


FIG. 5. Decay of intensity during heating of basalt with three magnetizations.

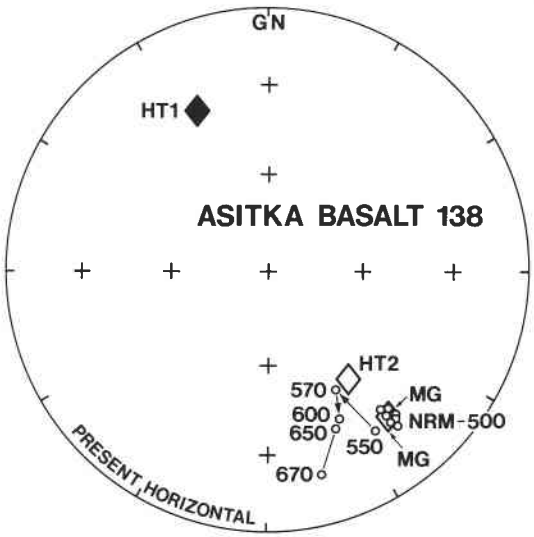


FIG. 6. Direction changes relative to present horizontal during heating of the basalt specimen with three magnetizations. The diamonds MG, HT1, and HT2 are the directions of the three magnetization.

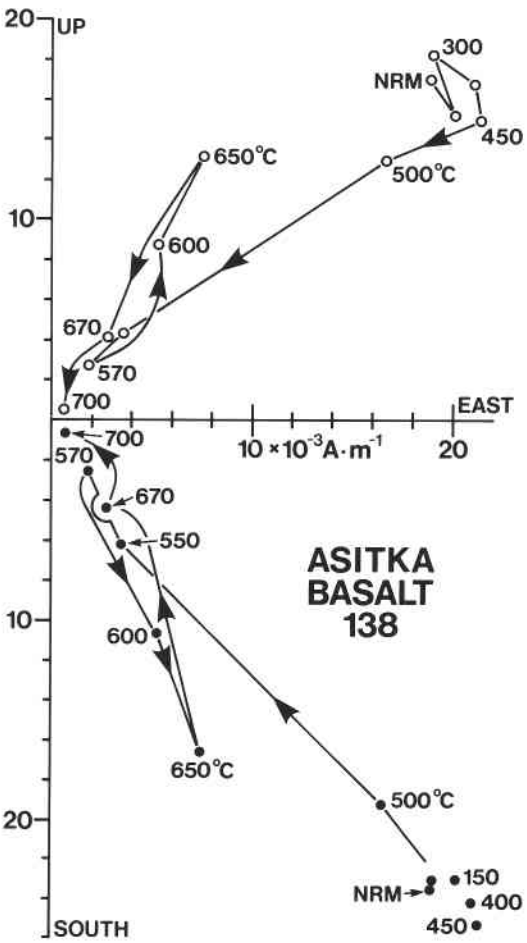


FIG. 7. Orthogonal diagrams for thermal demagnetization of the basalt with three magnetizations. Points are labelled in °C.

TABLE 1. Summary of results

Parameter	Rock				
	Tuff	Rhyolite	Basalt (direct)	Basalt[a] (indirect)	Basalt (combined)
D (deg)	354	85	132	126	129
I (deg)	40	−38	−43	−38	−40
S, N, n	5, 10, 15	5, 10,13	8, 8, 8	7, 8, 12	15, 16, 20
k	41	209	32	16	20
α_{95} (deg)	8	3	10	11	8
θ (deg)	56N	15N	—	—	40N
ϕ (deg)	63E	159E	—	—	123E
dm (deg)	9	4	12	13	9
dp (deg)	6	2	8	8	5
λ_p (deg)	23 ± 6	21 ± 4	—	—	23 ± 5
RPD (north) (deg)	4 ± 5	5 ± 4	—	—	4 ± 5
RPD (south) (deg)	50 ± 5	48 ± 4	—	—	49 ± 5
RR (north) (deg)	−45 ± 6	135 ± 5	—	—	180 ± 6
RR (south) (deg)	135 ± 6	44 ± 5			0 ± 6

NOTES: D and I are the declination and inclination of the mean; S is the number of oriented hand samples; N is the number of separate cores; n is the number of specimens (given unit weight to specimens in the calculations) studied; k is Fisher's precision; α_{95} is the radius of the circle of confidence; θ and ϕ are the latitude and longitude of the paleopole; dm and dp are the semiaxes of the error oval around the pole; and λ_p is the paleolatitude. RPD is the relative paleolatitudinal displacement (positive if displaced from the south) and RR the relative rotation (positive if anticlockwise relative to craton). RPD's and RR's are calculated using cratonic reference paleopole (for the Early Permian 43°N, 126°E, error (P = 0.05) A_{95} = 4° (Irving and Irving 1982)). RPD's and RR's for both northern and southern hemisphere options are given. Errors have been corrected by the method of Demarest (1983). Treatments are as follows: tuffs 550–650°C (9 specimens), 60–100 mT (6 specimens); rhyolite 550–600°C (11 specimens), 100 mT (2 specimens); basalt (direct) all specimens cleaned at 550°C; basalt (indirect, magnetization removed in the internals specified) 20–500°C (3 specimens), 570–650°C (1 specimen), 650–670°C (1 specimen), 670–700°C (1 specimen), 550–575°C (6 specimens). Bedding attitudes are as follows: tuff 14° at 138°E of geographic north; rhyolite 15° at 235°; basalt 40° at 230°.

[a]Includes one direction (HT1 of Fig. 5) with positive inclination, which has been inverted and averaged with the other.

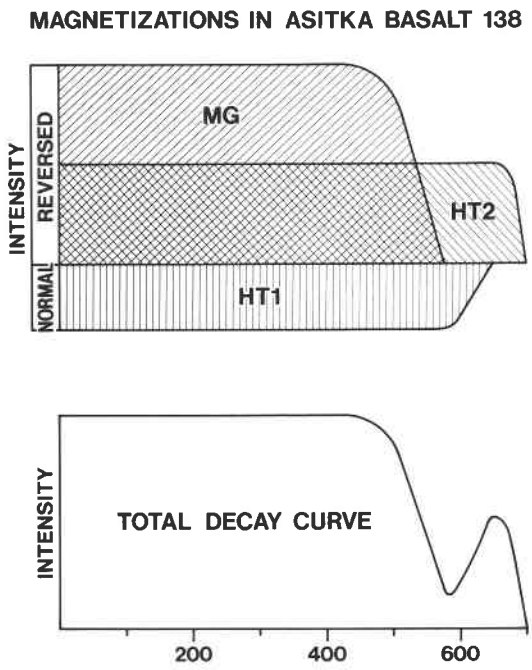


FIG. 8. Idealized unblocking curves for MG (magnetite) HT1 (low-temperature hematite), and HT2 (pure hematite) magnetizations, a possible model for the basalt with three magnetizations. Note that HT1 is antiparallel to MG and HT2.

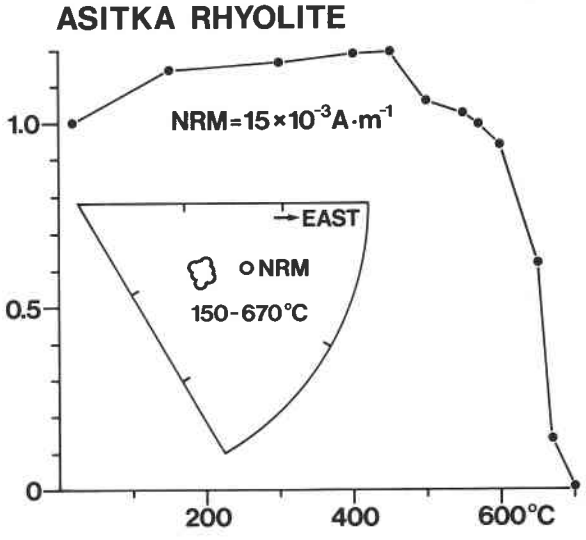


FIG. 9. Intensity and direction changes relative to present horizontal for rhyolite.

probability. There is, however, a southerly bias. In Fig. 13 this result is plotted on the paleolatitude map for the Early Permian derived from the cratonic reference paleopole (documented in the caption). This reference paleopole, based on data from

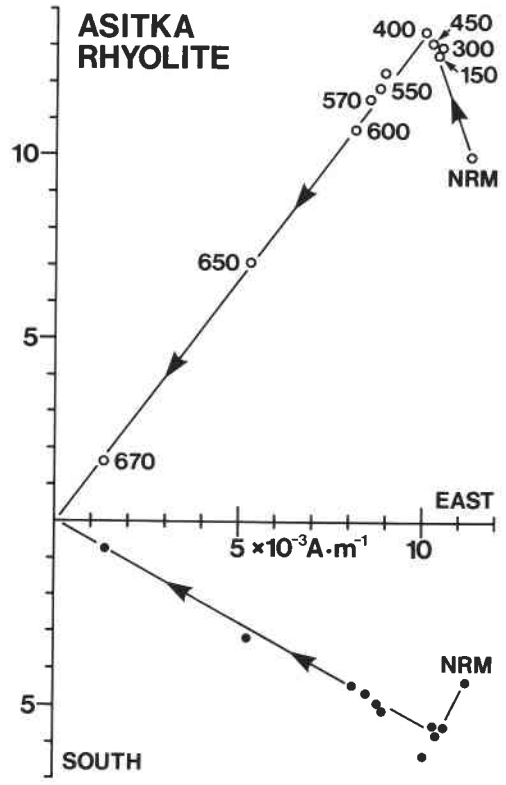


FIG. 10. Orthogonal diagrams for thermal demagnetization of rhyolite.

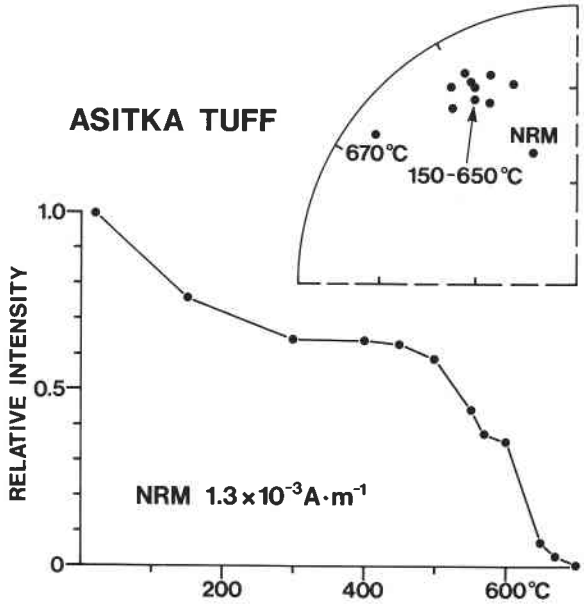


FIG. 11. Intensity and directional changes relative to present horizontal for tuff.

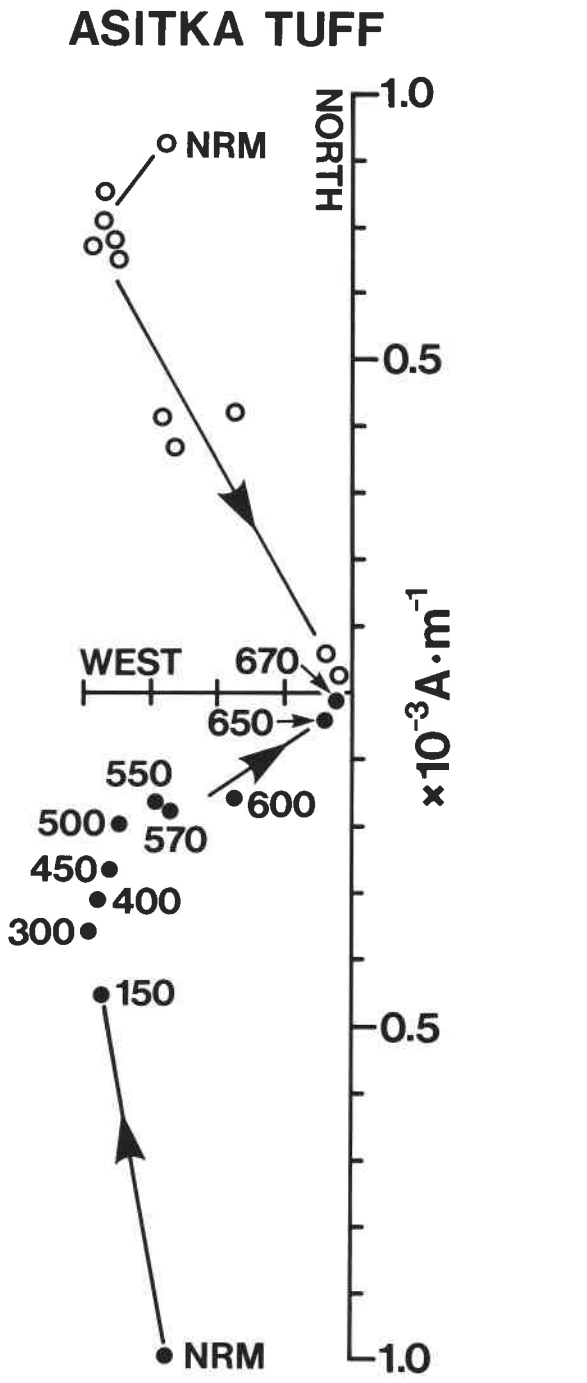


FIG. 12. Orthogonal diagrams for thermal demagnetization of tuff.

sedimentary rocks of central North America, has been confirmed by observations from basalt lavas of Ellesmere Island (Wynne *et al.* 1984). This means that the cratonic reference is not grossly in error from possible inclination errors in the Permian sedimentary rocks of the craton.

The data, although derived from highly stable magnetizations, are not numerous. It could be argued that the full range of the secular variation has not been averaged out, yet a considerable time span probably has been sampled, since both polarities are present. If the paleolatitude of the Asitka Group had been substantially different from that observed, it is unlikely that we would have obtained the approximate conformity to cratonic values of Fig. 13.

The declinations are widely scattered—a scatter that is unlikely to have been caused by secular variation, because the inclinations are in good agreement, and because the secular variation during the Kiaman is known to have been comparable to or somewhat smaller than that at present. The mean standard

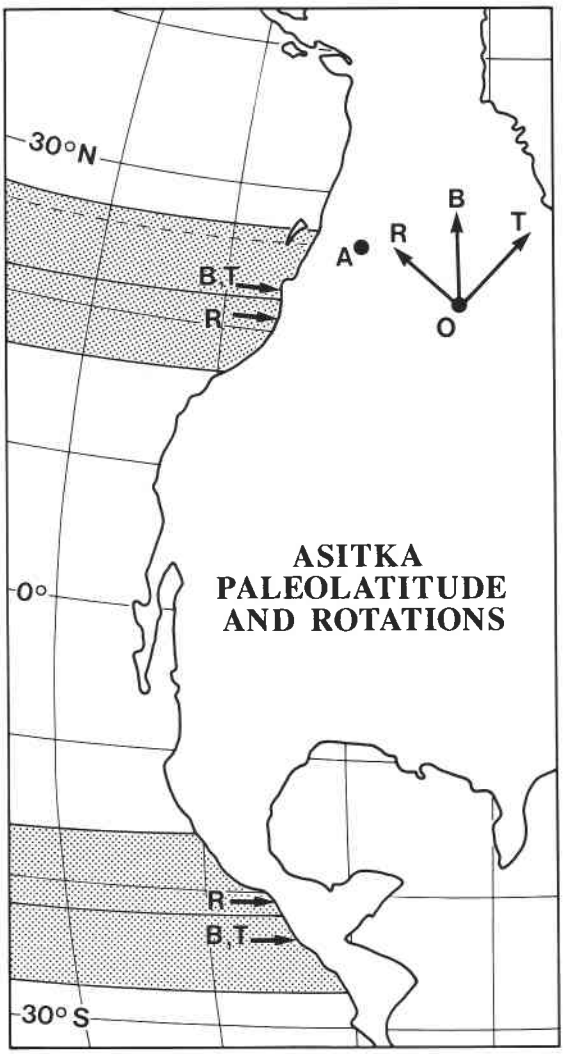


FIG. 13. Paleolatitude and rotations of Asitka Group. The 95% error zone in the paleolatitude determination is stippled. Both northern and southern options are shown. Individual paleolatitudes for tuff, rhyolite, and basalt are labelled T, R, and B. A is the sampling locality (dashed line is the expected paleolatitude). O is the observed mean paleolatitude. The arrows labelled T, R, and B are the horizontal components of the north-seeking directions of magnetization in tuff, rhyolite, and basalt respectively. Details in Table 1.

deviation of the Kiaman field at this paleolatitude would be expected to be about 13° (Irving and Pullaiah 1976), far less than the departures observed (Fig. 13). Evidently, the deformation has been such that the three sampling localities, which are separated by faults (Fig. 2), have rotated about vertical axes relative to one another; the rhyolite has rotated approximately 44° anticlockwise and the tuff about 47° clockwise relative to the basalt section.

The Permian fusulinid faunas from the west side of Stikinia are similar to those in the McCloud limestone of the eastern Klamath mountains of California. Paleomagnetic results from Permian rocks of the eastern Klamaths (mainly the Upper Permian Dekkas andesite that overlies the McCloud) obtained by Mankinen *et al.* (1982) are compared with the Asitka result in Fig. 14. Although the error is substantial, the means are in excellent agreement, which is consistent with the idea that

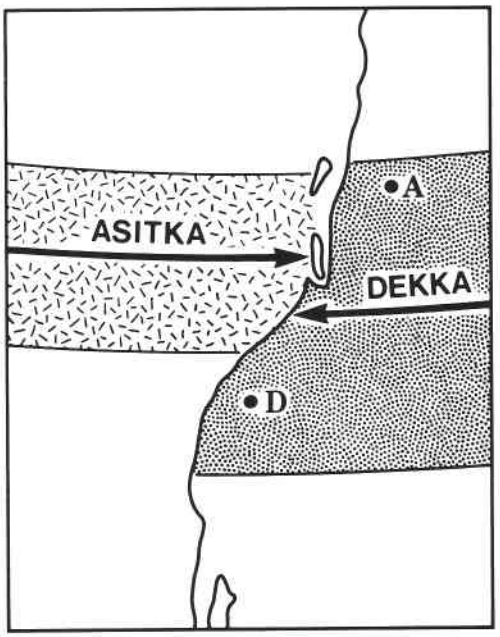


FIG. 14. Paleolatitudes from Stikinia (Lower Permian Asitka Group) and the eastern Klamaths (Upper Permian, mainly Dekkas andesite (Mankinen *et al.* 1982)) compared. The result from the Klamaths (paleopole 26°N, 324°E, error ($P = 0.05$) $A_{95} = 13°$) has been calculated using a Late Permian cratonic reference paleopole for North America (250 Ma, 49°N, 116°E, $A_{95} = 6°$ (Irving and Irving (1982)).

Stikinia and the eastern Klamaths were closer together in the Permian than they are at present.

Acknowledgments

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